Chapter 9 — Acid-Base Chemistry

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Bronsted acid is:

(a) Salt **(b)** Proton acceptor **(c)** Electron donor only **(d)** Proton donor

2. Bronsted base is:

(a) Proton donor **(b)** Electron acceptor only **(c)** Oxide only **(d)** Proton acceptor

3. Lewis acid is:

(a) Proton donor only **(b)** Electron-pair acceptor **(c)** Electron-pair donor **(d)** Salt

4. Lewis base is:

(a) Metal only **(b)** Electron-pair acceptor **(c)** Proton donor only **(d)** Electron-pair donor

5. Conjugate base of HCl:

(a) OH^- **(b)** H_2 **(c)** H_3O^+ **(d)** Cl^-

6. Water is amphoteric because:

(a) Only acid **(b)** Only base **(c)** Acts as acid and base **(d)** Neutral always

7. pH equals:

(a) $[H^+]$ **(b)** $-\log[OH^-]$ **(c)** $-\log[H^+]$ **(d)** $\log[H^+]$

8. pOH equals:

(a) $-\log[OH^-]$ **(b)** $[OH^-]$ **(c)** $\log[H^+]$ **(d)** $-\log[H^+]$

9. Kw at 298 K:

(a) 14 **(b)** 1.0×10^{-7} **(c)** 7 **(d)** 1.0×10^{-14}

10. pH + pOH at 298 K:

(a) 1 **(b)** 14 **(c)** 0 **(d)** 7

11. 0.01 M HCl pH:

(a) 12 **(b)** 2 **(c)** 7 **(d)** 1

12. 0.01 M NaOH pH:

(a) 1 **(b)** 2 **(c)** 7 **(d)** 12

13. Ka measures:

(a) Weak acid strength **(b)** Solubility only **(c)** Base strength only **(d)** Rate

14. Larger Ka means:

(a) No ionization **(b)** Neutral salt **(c)** Weaker acid **(d)** Stronger acid

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15. Acidic buffer example:

(a) CH_3COOH/CH_3COONa **(b)** NaOH/NaCl **(c)** HCl/NaCl **(d)** H_2O/NaCl

16. Basic buffer example:

(a) CH_3COOH/NaCl **(b)** HCl/NH_4Cl **(c)** NH_3/NH_4Cl **(d)** NaOH/NaCl

17. Buffer resists:

(a) Colour change only **(b)** pH change **(c)** Temperature change **(d)** Mass change

18. Common ion effect suppresses:

(a) Boiling **(b)** Precipitation always **(c)** Ionization **(d)** Combustion

19. CH_3COONa solution is:

(a) Amphoteric **(b)** Basic **(c)** Acidic **(d)** Neutral

20. NH_4Cl solution is:

(a) Acidic **(b)** Basic **(c)** Amphoteric **(d)** Neutral

21. NaCl solution is:

(a) Acidic **(b)** Buffer **(c)** Basic **(d)** Neutral

22. Phenolphthalein in base is:

(a) Colourless **(b)** Red **(c)** Yellow **(d)** Pink

23. Methyl orange in acid is:

(a) Colourless **(b)** Yellow **(c)** Blue **(d)** Red

24. Weak acid-strong base titration uses:

(a) Methyl orange only **(b)** No indicator **(c)** Phenolphthalein **(d)** Litmus only

25. Strong acid-weak base titration uses:

(a) Phenolphthalein only **(b)** No indicator **(c)** Methyl orange **(d)** Starch only

26. Ksp predicts:

(a) Colour **(b)** Rate only **(c)** pH only **(d)** Precipitation

27. BF_3 is:

(a) Bronsted acid **(b)** Salt **(c)** Lewis base **(d)** Lewis acid

28. NH_3 is:

(a) Salt **(b)** Lewis base **(c)** Lewis acid **(d)** Bronsted acid only

29. Strong acid ionizes:

(a) Only in base **(b)** Not at all **(c)** Partially **(d)** Completely

30. Neutralization produces:

(a) Only base **(b)** Salt and water **(c)** Only acid **(d)** Only gas

31. Arrhenius acid produces:

(a) Neutrons **(b)** Electrons **(c)** OH^- in water **(d)** H^+ in water

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32. Bronsted acid is:

(a) Salt **(b)** Proton acceptor **(c)** Proton donor **(d)** Electron donor

33. Lewis base is:

(a) Proton donor only **(b)** Metal only **(c)** Electron-pair acceptor **(d)** Electron-pair donor

34. Conjugate base of NH_4^+ is:

(a) NH_3 **(b)** NH_2^- **(c)** N_2 **(d)** H_3O^+

35. Kw equals:

(a) $[H^+] + [OH^-]$ **(b)** $[H^+][OH^-]$ **(c)** $[H^+]/[OH^-]$ **(d)** $[OH^-]/[H^+]$

36. pH of 0.001 M HCl is:

(a) 11 **(b)** 3 **(c)** 7 **(d)** 1

37. pOH of 0.001 M NaOH is:

(a) 11 **(b)** 3 **(c)** 7 **(d)** 1

38. Strong acid ionizes:

(a) Only in base **(b)** Completely **(c)** Not at all **(d)** Partially

39. Weak acid ionizes:

(a) Partially **(b)** Only in base **(c)** Not at all **(d)** Completely

40. Larger Ka means:

(a) No ionization **(b)** Neutral salt **(c)** Weaker acid **(d)** Stronger acid

41. pKa equals:

(a) log Ka **(b)** Ka/10 **(c)** -log Ka **(d)** 10/Ka

42. Common ion effect suppresses:

(a) Redox **(b)** Ionization **(c)** Boiling **(d)** Combustion

43. Acidic buffer contains:

(a) Water only **(b)** Strong base and salt **(c)** Weak acid and its salt **(d)** Strong acid and salt

44. Basic buffer contains:

(a) Strong acid and salt **(b)** Weak base and its salt **(c)** Strong base and salt **(d)** Water only

45. Henderson-Hasselbalch equation gives:

(a) Rate constant **(b)** Gas volume **(c)** Bond order **(d)** Buffer pH

46. Ksp expression for AgCl is:

(a) $[Cl^-]/[Ag^+]$ **(b)** $[Ag^+][Cl^-]$ **(c)** $[Ag^+]/[Cl^-]$ **(d)** $[AgCl]$

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47. Precipitation occurs when ionic product:

(a) Exceeds Ksp **(b)** Is below Ksp **(c)** Equals zero **(d)** Is always one

48. Salt of strong acid and strong base is:

(a) Neutral **(b)** Basic **(c)** Buffer **(d)** Acidic

49. Phenolphthalein in acid is:

(a) Red **(b)** Pink **(c)** Blue **(d)** Colourless

50. Equivalence point is:

(a) Always pH 7 **(b)** No reaction **(c)** Stoichiometric neutralization **(d)** Indicator colour only

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Answer Key with Brief Reasons

1. Answer: (d) Proton donor — Acid-base concept.

2. Answer: (d) Proton acceptor — Acid-base concept.

3. Answer: (b) Electron-pair acceptor — Lewis concept.

4. Answer: (d) Electron-pair donor — Lewis concept.

5. Answer: (d) Cl^- — Remove H^+.

6. Answer: (c) Acts as acid and base — Donates/accepts H^+.

7. Answer: (c) -log[H^+] — Definition.

8. Answer: (a) -log[OH^-] — Definition.

9. Answer: (d) 1.0×10^{-14} — Ionic product.

10. Answer: (b) 14 — From Kw.

11. Answer: (b) 2 — Strong acid fully ionized.

12. Answer: (d) 12 — pOH 2, pH 12.

13. Answer: (a) Weak acid strength — Ionization constant.

14. Answer: (d) Stronger acid — More ionization.

15. Answer: (a) CH_3COOH/CH_3COONa — Weak acid + salt.

16. Answer: (c) NH_3/NH_4Cl — Weak base + salt.

17. Answer: (b) pH change — Buffer action.

18. Answer: (c) Ionization — Le Chatelier.

19. Answer: (b) Basic — Salt hydrolysis.

20. Answer: (a) Acidic — Salt hydrolysis.

21. Answer: (d) Neutral — Strong acid/base salt.

22. Answer: (d) Pink — Indicator colour.

23. Answer: (d) Red — Indicator colour.

24. Answer: (c) Phenolphthalein — End point in basic range.

25. Answer: (c) Methyl orange — End point in acidic range.

26. Answer: (d) Precipitation — Solubility equilibrium.

27. Answer: (d) Lewis acid — Accepts lone pair.

28. Answer: (b) Lewis base — Donates lone pair.

29. Answer: (d) Completely — Strong electrolyte.

30. Answer: (b) Salt and water — Acid + base.

31. Answer: (d) H^+ in water — Aqueous definition.

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32. Answer: (c) Proton donor — Proton transfer.

33. Answer: (d) Electron-pair donor — Lone pair donation.

34. Answer: (a) NH_3 — Remove H^+.

35. Answer: (b) $[H^+][OH^-]$ — Ionic product.

36. Answer: (b) 3 — Strong acid.

37. Answer: (b) 3 — Strong base.

38. Answer: (b) Completely — Strong electrolyte.

39. Answer: (a) Partially — Equilibrium.

40. Answer: (d) Stronger acid — More ionization.

41. Answer: (c) -log Ka — Logarithmic scale.

42. Answer: (b) Ionization — Le Chatelier.

43. Answer: (c) Weak acid and its salt — Buffer pair.

44. Answer: (b) Weak base and its salt — Buffer pair.

45. Answer: (d) Buffer pH — pH=pKa+log salt/acid.

46. Answer: (b) $[Ag^+][Cl^-]$ — Sparingly soluble salt.

47. Answer: (a) Exceeds Ksp — Supersaturation.

48. Answer: (a) Neutral — No hydrolysis.

49. Answer: (d) Colourless — Indicator range.

50. Answer: (c) Stoichiometric neutralization — Mole ratio point.

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